

COMPUTER NETWORK DESIGN

**A working network design where
devices can communicate correctly
using common networking protocols
and services**

Done By:

Diya Basnet

ID: 11907776

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Overview

This project is a small virtual computer network that I designed, configured, and tested in Cisco Packet Tracer. The goal was to build a network where client devices can actually communicate with a server using the core protocols that run real networks, and then to capture that traffic and show how each protocol behaves.

The network is built around a single internal LAN that connects to a simulated external network. An edge router acts as the gateway and handles DHCP, NAT, and routing. A switch ties all of the internal devices together, and two servers provide DNS and web services. A second router stands in for the internet so I can show traffic moving between the private network and the outside world.

I used Packet Tracer's built-in Simulation Mode to capture and inspect packets. It captures the exact traffic on the simulated topology and lets me step through each exchange packet by packet, which made it easy to show protocols like DHCP, ARP, DNS, TCP, and UDP working on the wire.

Objectives

The objectives of the project are:

- Design a network that includes a router, a switch, multiple client devices, and servers, with a simulated connection to an external network.
- Assign IP addressing using a structured private scheme and confirm every device can reach the others.
- Configure the core network services like DHCP for automatic addressing, DNS for name resolution, and a web server for HTTP and HTTPS
- Demonstrate NAT by translating private internal addresses to a single public address
- Capture and analyze traffic to show DHCP, DNS, ARP, ICMP, TCP, UDP, and SSL/TLS actually working

Network Design Explanation

The topology is one internal LAN connected through an edge router to a simulated external network. Router0 (a Cisco ISR 4321) sits at the edge and acts as the gateway for the whole LAN. Its inside interface (Gig0/0/0) holds the address 192.168.1.1 and faces the internal network through switch. Its outside

interface (Gig0/0/1) holds 203.0.113.1 and connects to Router 1, which represents the internet.

Switch0 (A Cisco 2960) connects all of the internal devices which are two PCs, two laptops, a Web Server, and DNS server. The client PCs and laptops receive their addressing automatically through DHCP, while the two servers use static addresses so they can always be found at a fixed location. The Web Server lives at 192.168.1.10 and the DNS Server at 192.168.1.20.

The internal LAN uses the private block 192.168.1.0/24, which gives plenty of usable host addresses and keeps a clean logical boundary for the whole network. The link between the two routers uses a small 203.0.113.0/30 subnet, which is the standard way to address a point-to-point link between two routers because it only needs two usable addresses.

Device	IP Address	Subnet	Gateway	Assignment
Router0 (LAN - Gig0/0/0)	192.168.1.1	255.255.255.0		Static
Router0 (WAN - Gig0/0/1)	203.0.113.1	255.255.255.252		Static
Router1(Gig0/0)	203.0.113.2	255.255.255.252		Static
Web Server	192.168.1.10	255.255.255.0	192.168.1.1	Static
DNS Server	192.168.1.20	255.255.255.0	192.168.1.1	Static
PC0	192.168.1.52	255.255.255.0	192.168.1.1	DHCP
PC1	192.168.1.51	255.255.255.0	192.168.1.1	DHCP
Laptop 0 & 1	192.168.1.53	255.255.255.0	192.168.1.1	DHCP

Table 1: IP ADDRESSING TABLE

The diagram below shows the full network layout, including the two routers, the switch, all client devices, both servers, and the addressing for each connection:

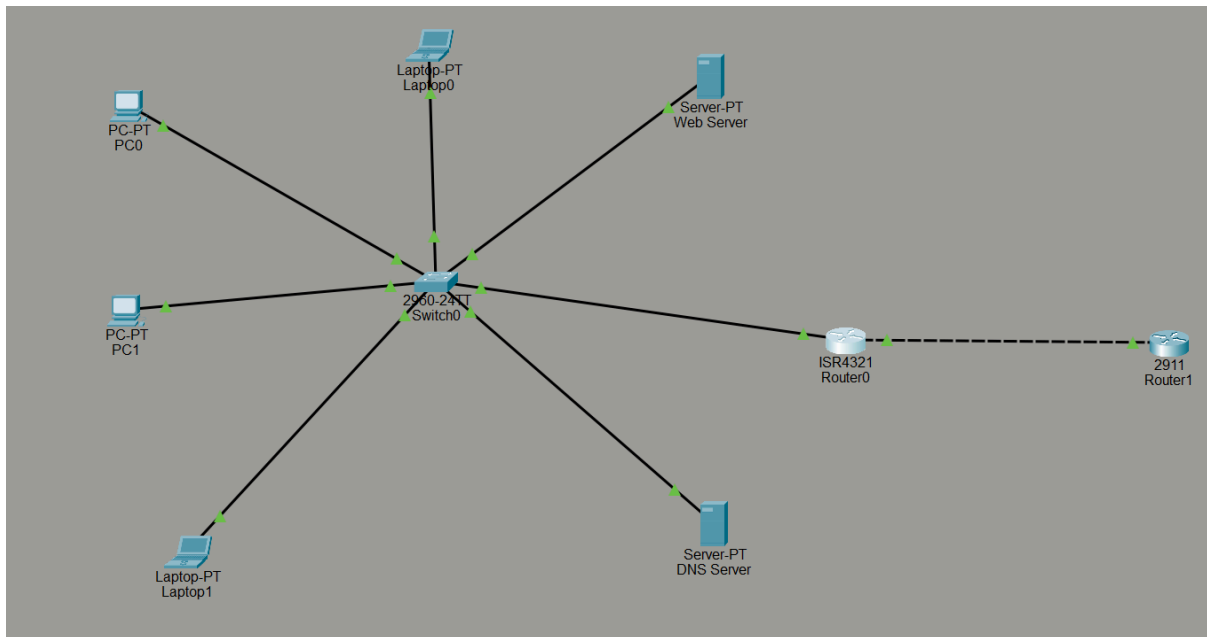


Figure 1: Network Topology Diagram

Networking Protocols

The network demonstrates ten core networking protocols and services. Each one plays a specific role, and several of them work together on a single action. For example, browsing a web page by name uses DNS, then ARP, then a TCP handshake, then HTTP.

The table below summarizes what each protocol does in this network:

Protocol	Role
DHCP	Automatically assigns IP addresses, subnet mask, gateway, and DNS to the client PCs and laptops from a pool on Router0.
DNS	Resolves the domain name www.diyagnet.com to the Web Server's IP address (192.168.1.10) so clients can browse by name instead of number.
ARP	Maps a known IP address to its MAC address on the local network before any data is sent at Layer 2.
HTTP	Delivers the web page from the Web Server to the client browser over an unencrypted TCP connection (port 80).
HTTPS/SSL/TLS	The secure version of HTTP. SSL/TLS negotiates an encrypted session (port 443) before the page is transferred.

ICMP	Used by ping and traceroute to test reachability and trace the path packets take between devices.
NAT	Translates the private 192.168.1.x source addresses into the single public address 203.0.113.1 when traffic leaves for the external network.
TCP	Connection-oriented, reliable transport, used by HTTP and HTTPS. It establishes a session with a three-way handshake (SYN, SYN-ACK, ACK).
UDP	Connectionless, lightweight transport. It carries DHCP and DNS traffic where speed matters more than guaranteed delivery.
IP/ SUBNETTING	The internal LAN uses private block 192.168.1.0/24. The WAN link between the two routers uses a small 203.0.113.0/30 subnet.

Table 2: PROTOCOLS TABLE

Device Configuration

All router configuration was done through the CLI, and the servers were configured through their Config and Services tab.

- **Router0 Interface Configuration**

Both interfaces on Router0 were given IP addresses and brought up with the no shutdown command. Gig0/0/0 is the LAN gateway (192.168.1.1) and Gig0/0/1 is the WAN link (203.0.113.1)

```

Router0
Physical Config CLI Attributes
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: yes

At any point you may enter a question mark '?' for help.
Use ctrl-c to abort configuration dialog at any prompt.
Default settings are in square brackets '[]'.

Basic management setup configures only enough connectivity
for management of the system, extended setup will ask you
to configure each interface on the system

Would you like to enter basic management setup? [yes/no]: enable
% Please answer 'yes' or 'no'.
Would you like to enter basic management setup? [yes/no]: yes
Configuring global parameters:

Enter host name [Router]:

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig 0/0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up
exit
Router(config)#interface gig0/0/1
Router(config-if)#ip address 203.0.113.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
exit
Router(config)#

```

Figure 2: Router0 Interface Configuration

- **DHCP Configuration**

A DHCP pool named LAN was created on Router0. It hands out addresses on the 192.168.1.0/24 network, sets the default gateway to 192.168.1.1, and points clients to the DNS Server at 192.168.1.20. The first fifty addresses are excluded so they stay reserved for static devices like the servers and router.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.50
Router(config)#ip dhcp pool LAN
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 192.168.1.20
Router(dhcp-config)#exit
```

Figure 3: DHCP pool configuration on Router0

- **NAT Configuration**

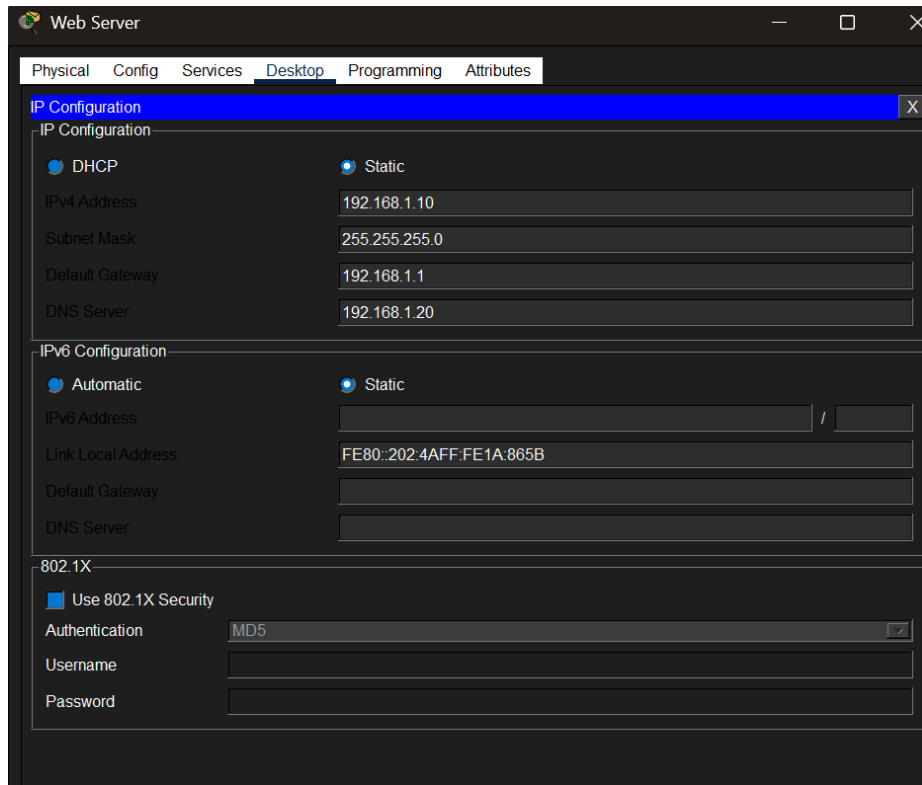
NAT was configured so the whole internal network can share the single public address on the WAN interface. Gig0/0/0 was marked as the NAT inside interface and Gig0/0/1 as the outside. An access list identifies the internal 192.168.1.0/24 addresses, and the overload keyword enables PAT so many hosts can share one public IP.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#interface gig0/0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat inside source list 1 interface gig0/0/1 overload
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 4: NAT configuration on Router0

- **Server Static IP Configuration**

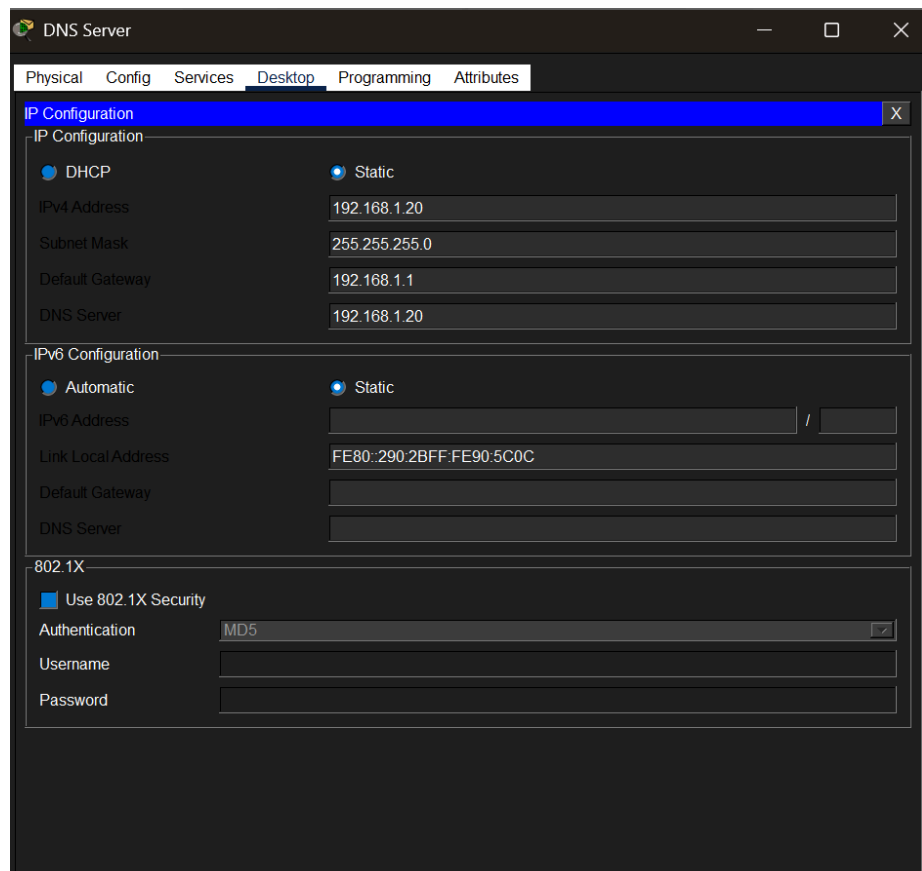
The Web Server and DNS were given static addresses so they were always at a known location. The Web Server is at 192.168.1.10 and the DNS Server is at 192.168.1.20; both use 192.168.1.1 as their gateway and point to the DNS Server for name resolution.



The image shows a 'Web Server' configuration window with tabs for Physical, Config, Services, Desktop, Programming, and Attributes. The 'IP Configuration' tab is active, showing settings for IP Configuration, IPv6 Configuration, and 802.1X.

Section	Option	Value
IP Configuration	DHCP	☐ DHCP
	Static	☒ Static
	IPv4 Address	192.168.1.10
	Subnet Mask	255.255.255.0
	Default Gateway	192.168.1.1
	DNS Server	192.168.1.20
IPv6 Configuration	Automatic	☐ Automatic
	Static	☒ Static
	IPv6 Address	
	Link Local Address	FE80::202:4AFF:FE1A:865B
	Default Gateway	
	DNS Server	
802.1X	Use 802.1X Security	☐ Use 802.1X Security
	Authentication	MD5
	Username	
	Password	

Figure 5: Web Server static IP Configuration



The image shows a 'DNS Server' configuration window with tabs for Physical, Config, Services, Desktop, Programming, and Attributes. The 'IP Configuration' tab is active, showing settings for IP Configuration, IPv6 Configuration, and 802.1X.

Section	Option	Value
IP Configuration	DHCP	☐ DHCP
	Static	☒ Static
	IPv4 Address	192.168.1.20
	Subnet Mask	255.255.255.0
	Default Gateway	192.168.1.1
	DNS Server	192.168.1.20
IPv6 Configuration	Automatic	☐ Automatic
	Static	☒ Static
	IPv6 Address	
	Link Local Address	FE80::290:2BFF:FE90:5C0C
	Default Gateway	
	DNS Server	
802.1X	Use 802.1X Security	☐ Use 802.1X Security
	Authentication	MD5
	Username	
	Password	

Figure 6: DNS Server Static IP Configuration

- **DNS Service Configuration**

On the DNS Server, the DNS service was turned on and an A record was added mapping the domain name www.diyagnet.com to the Web Server's IP address, 192.168.1.10. This is what lets clients reach the web page by name instead of by IP.

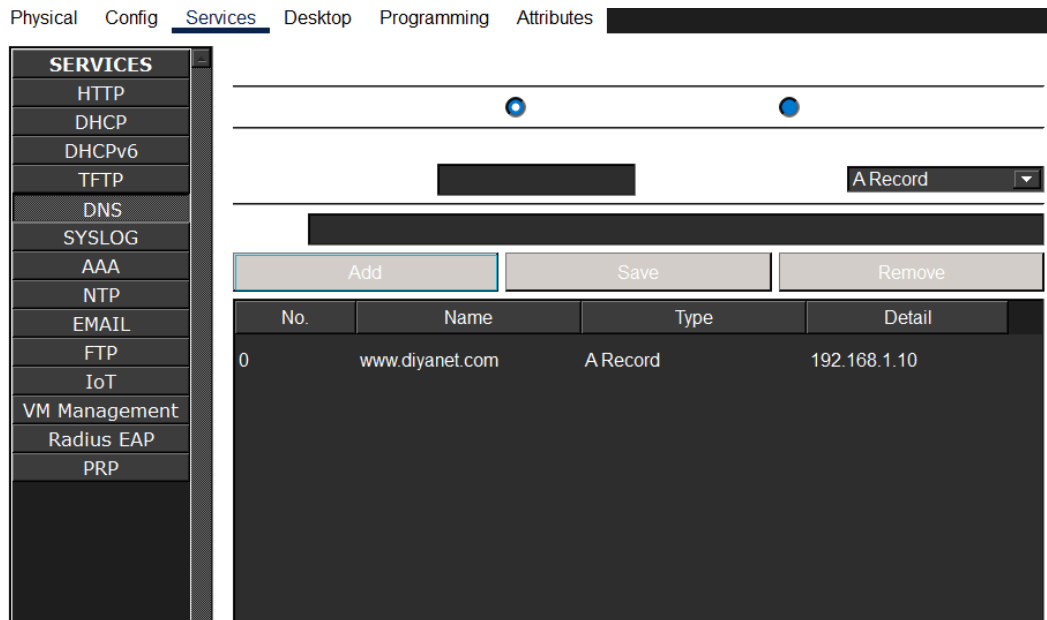


Figure 7: DNS A record mapping www.diyagnet.com to 192.168.1.10

Network Testing and Results

After configuration, the network was tested by bringing up each service and capturing the traffic in Simulation Mode.

- **DHCP - Automatic IP Assignment**

Each client was set to DHCP and successfully pulled an address from the pool. PC0 received 192.168.1.52 along with the correct subnet mask, gateway, and DNS server, and the interface reported “DHCP request successful”. This confirms the DORA exchange (Discover, Offer, Request, Acknowledge) completed correctly.

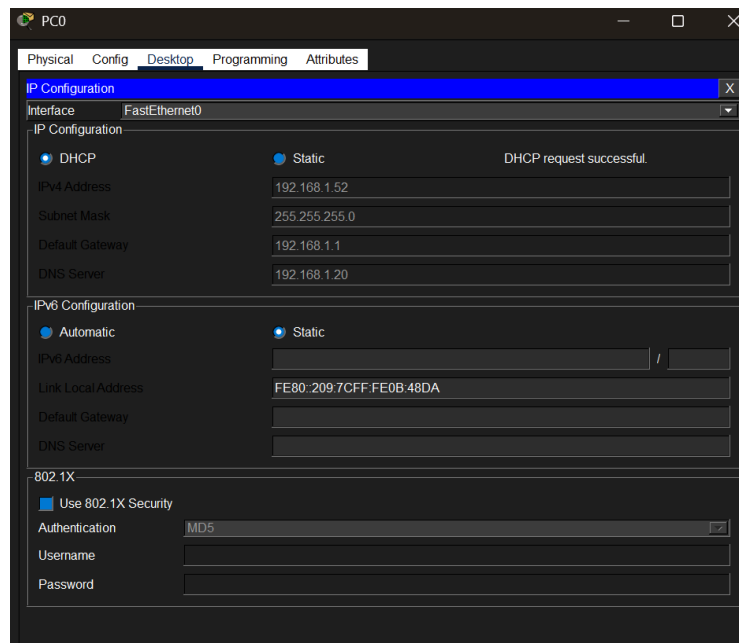


Figure 8: DHCP automatic IP assignment on PC0

• ARP and ICMP - Connectivity Test

To test connectivity, I cleared the ARP cache with `arp -d` and pinged the Web Server at 192.168.1.10. The ping returned four replies with 0% loss. The Event List on the right shows the ARP exchange happening first. The request floods out from the switch to every device, the Web Server replies, and only then do the ICMP echo packets flow. This test demonstrates ARP (resolving the MAC address) and ICMP (testing reachability).

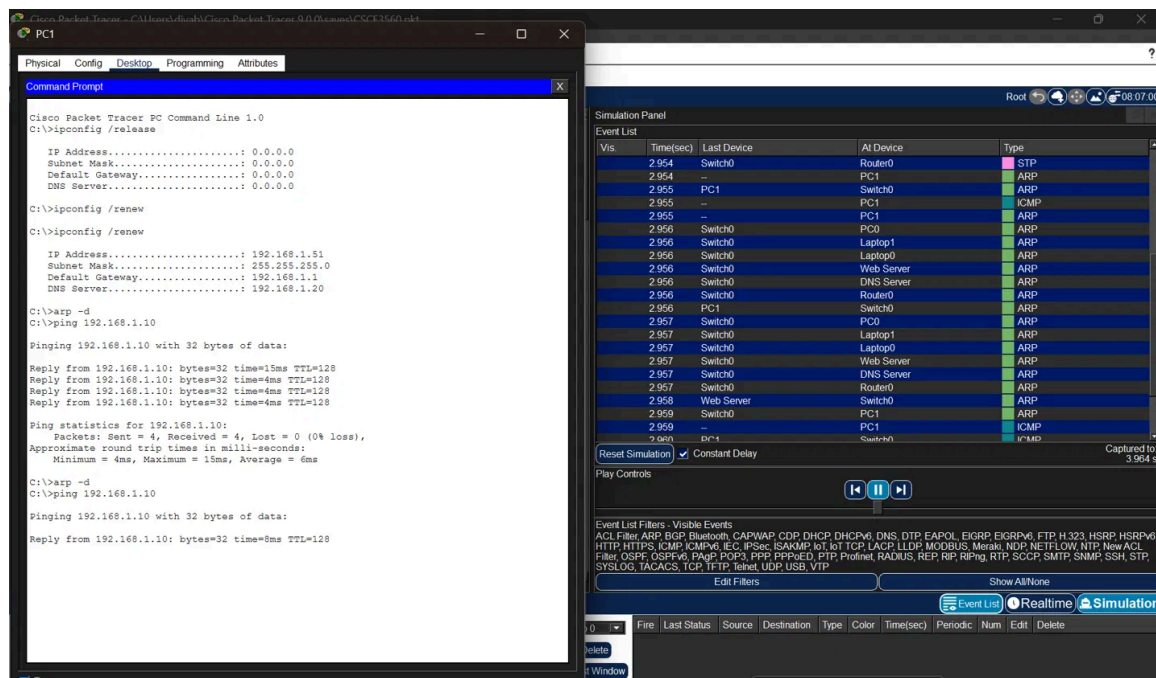


Figure 9: ARP resolution and ICMP ping test

- **ICMP - Traceroute**

I also ran a traceroute from a client to the external address 203.0.113.2. The trace shows the packet passing through the gateway at 192.168.1.1 on its way to the external network, confirming the routing path between the internal LAN and the outside.

```
C:\>tracert 203.0.113.2

Tracing route to 203.0.113.2 over a maximum of 30 hops:

  1    8 ms    4 ms    4 ms    192.168.1.1
```

Figure 10: Traceroute to the external network

- **DNS - Name Resolution**

Running nslookup www.diyagnet.com returned the address 192.168.1.10, which means the DNS Server correctly resolved the domain name to the Web Server. The Event List shows the DNS query and response travelling between the client and the DNS Server. Because DNS rides on UDP, this capture also demonstrates UDP traffic.

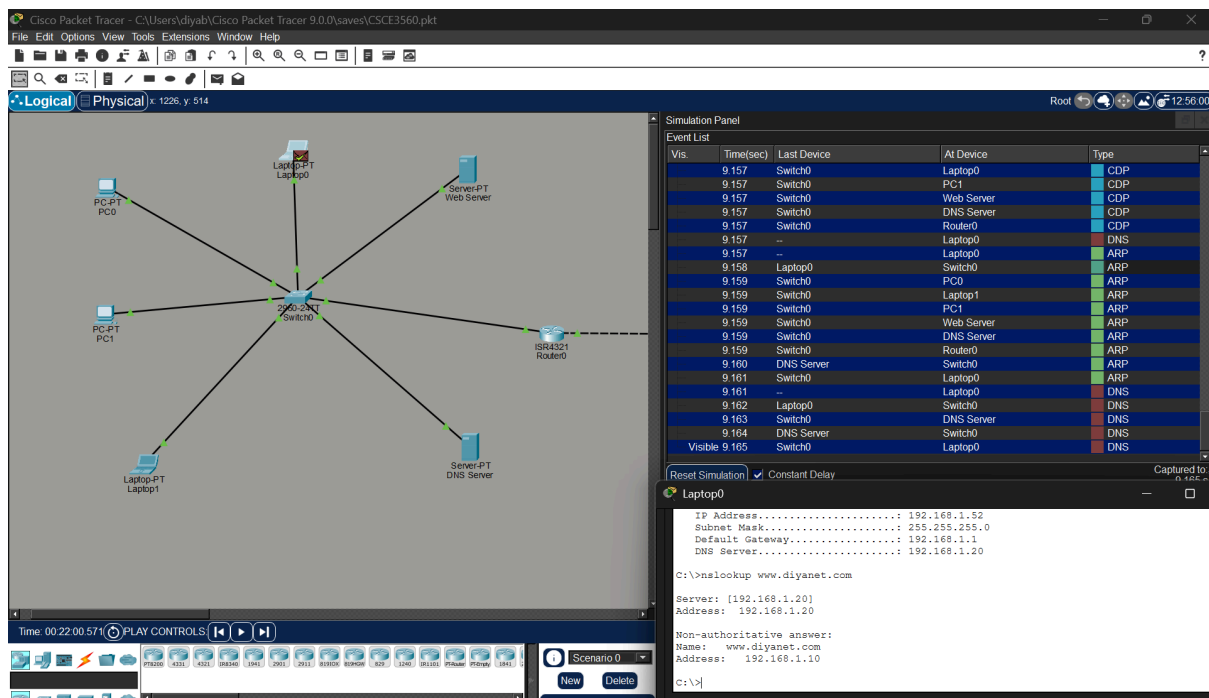


Figure 11: DNS name resolution

- **HTTP and TCP - Web Access**

From a client browser I opened <http://www.diyagnet.com> and the Web Server's page loaded successfully. The Event List shows TCP packets forming the three-way handshake (SYN, SYN-ACK, ACK) before the HTTP request is sent. This test demonstrates both HTTP (The web request) and TCP (the reliable connection underneath it).

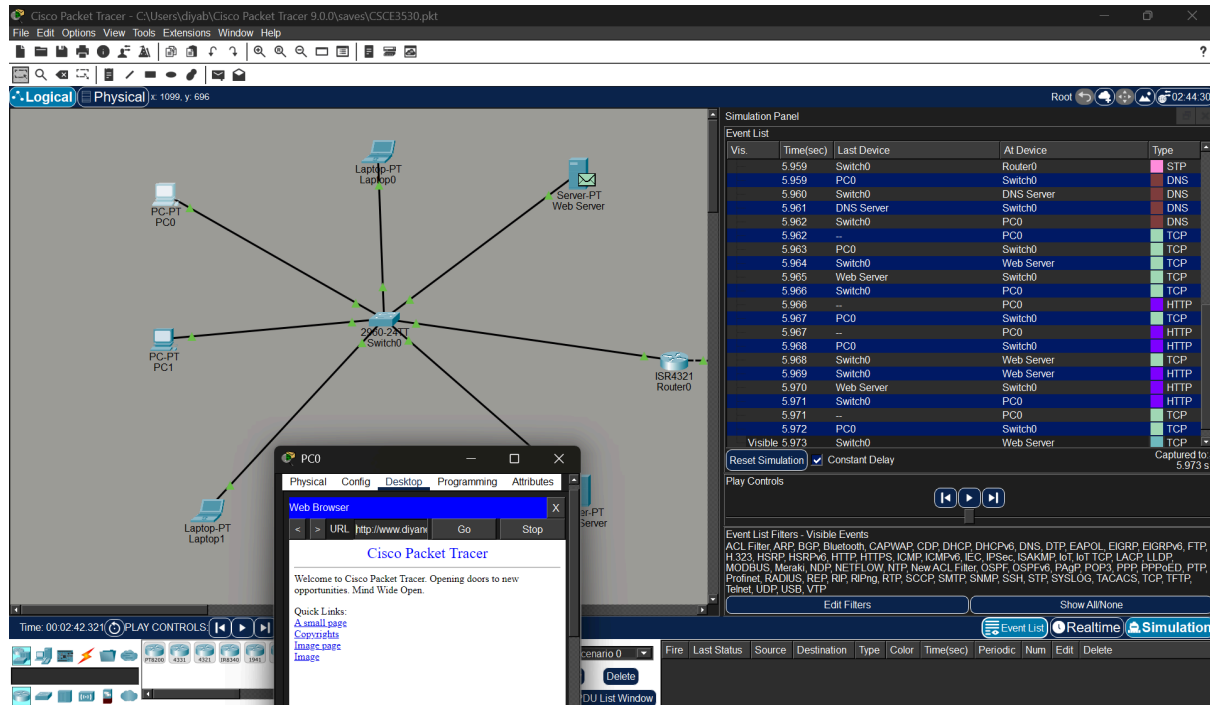


Figure 12: HTTP Web Access

- **HTTPS and SSL/TLS - Secure Web Access**

I then browsed <https://www.diyagnet.com> to test secure communication. The page is loaded over HTTPS, and the Event List shows HTTPS and TCP traffic where SSL/TLS negotiates an encrypted session before any page data is sent. This is the secure, encrypted version of the HTTP.

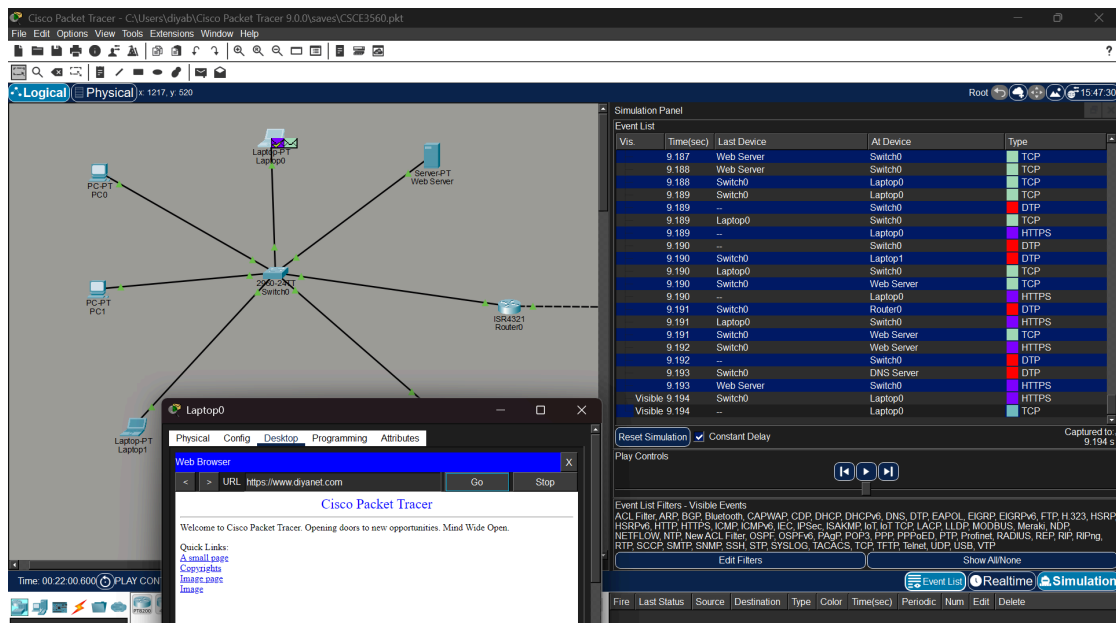


Figure 13: HTTPS/SSL -TLS Secure Web Access

- **NAT - Private to Public Translation**

Finally, after clients sent traffic toward the external network. I ran show ip nat translations on Router0. The translation table shows the internal 192.168.1.x source addresses being mapped to the single outside address 203.0.113.1. This proves NAT is working, multiple private hosts are sharing one public address to reach the external network.

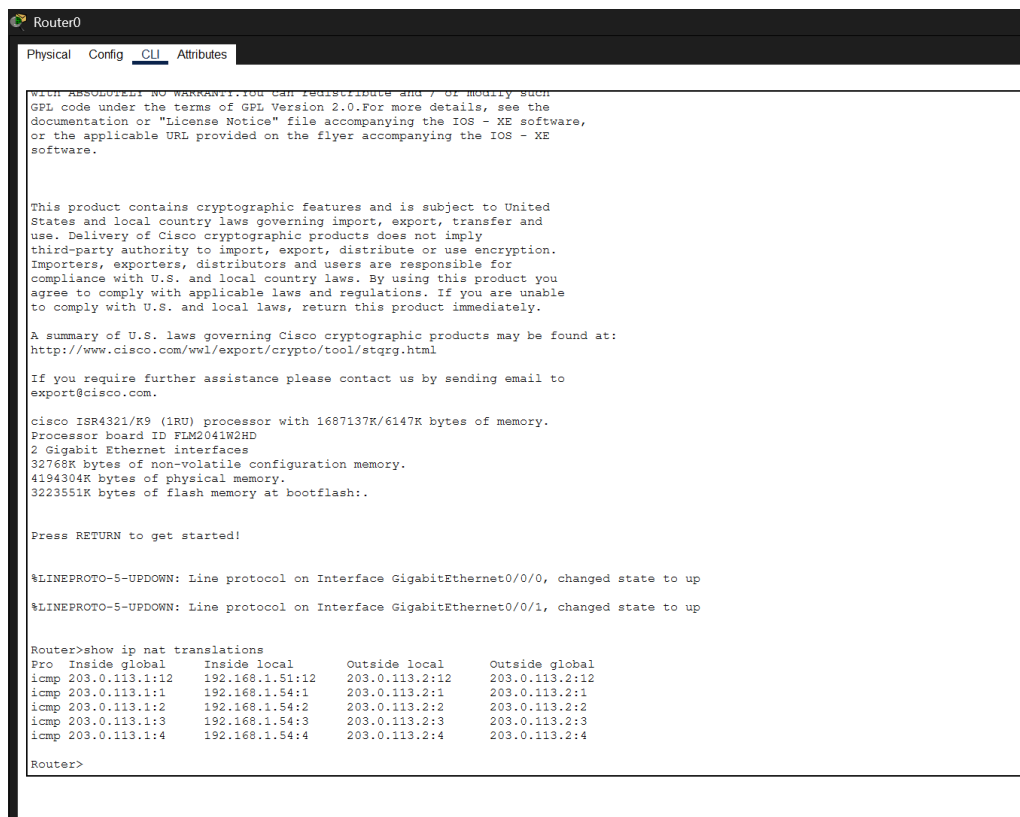


Figure 14: NAT translation table

Challenges Faced and Solutions

Some problems faced while building the networks are:

- **DHCP requests failing on the clients**

The biggest issue was that the client PCs kept reporting “DHCP request failed” even though the DHCP pool was configured on the router. After checking the router with `show ip interface` (which confirmed the interfaces were up) and `show running-config` (which confirmed the pool existed), I traced the problem to the cabling. The switch was connected to the wrong router port. It was plugged into the WAN interface (Gig0/0/1) instead of the LAN gateway interface (Gig0/0/0), so the DHCP requests never reached the interface that hands out addresses. Deleting that cable and reconnecting the switch to Gig0/0/0 fixed it immediately, and every client pulled an address.

- **Getting into the right CLI mode**

Early on, my configuration commands were rejected with “invalid input detected.” The problem was that I was typing configuration commands from user mode instead of configuration mode. Once I understood the prompt levels, typing `enable` to reach Router#, then `configure terminal` to reach Router(config)#. The commands worked.

- **Capturing specific protocols in Simulation Mode**

When I filtered the Event List to show only specific protocols like only DHCP, no packets appeared which was confusing. It had to do with the packet tracer filter so I switched to showing all the protocols at once and just tracked what I needed to test.

Conclusion

The project resulted in a working client-server network where every device can communicate correctly and every required protocol is demonstrated. Clients receive addresses automatically through DHCP, resolve names through DNS, reach a web server over both HTTP and secure HTTPS, and connect to an external network through NAT. Connectivity was verified with ping and traceroute, and each protocol DHCP, DNS, ARP, ICMP, HTTP, HTTPS, TCP, UDP & NAT was captured and analyzed in Simulation Mode.

Beyond just getting it working, the debugging made the concepts more clear. Tracing the DHCP failure back to a mislabeled switch port forced me to understand how DHCP requests actually travel to the gateway, and learning the CLI prompt levels made router configuration much less intimidating. By the end, I had a clear picture of how routing, switching, addressing, and application services all fit together to make a network function.